

Solution 1

Solution 1 (a)

The probability density function (PDF) for a d -dimensional multivariate Gaussian distribution $N(\mu, \Sigma)$ is:

$$f(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right)$$

Given a spherical Gaussian $N(0, I_d)$:

$$\mu = 0 \quad \text{and} \quad \Sigma = I_d \rightarrow (\text{definition: } N(\mu, \Sigma))$$

The determinant of any identity matrix is 1, and inverse of an identity matrix is itself:

$$|\Sigma| = |I_d| = 1 \quad \text{and} \quad \Sigma^{-1} = I_d^{-1} = I_d$$

Now, substitute μ , Σ , and $|\Sigma|$ into the PDF formula:

$$\begin{aligned} f(x) &= \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right) \\ &= \frac{1}{(2\pi)^{d/2} (1)^{1/2}} \exp \left(-\frac{1}{2} (x - \mathbf{0})^\top I_d (x - \mathbf{0}) \right) \\ &= \frac{1}{(2\pi)^{d/2}} \exp \left(-\frac{1}{2} x^\top x \right) \\ &= (2\pi)^{-d/2} \exp \left(-\frac{1}{2} \|x\|^2 \right) \end{aligned}$$

Next, simplify the expression by taking natural log such that $L(x) = \ln(f(x))$:

$$\begin{aligned} L(x) &= \ln \left((2\pi)^{-d/2} \exp \left(-\frac{1}{2} \|x\|^2 \right) \right) \\ &= \ln \left((2\pi)^{-d/2} \right) + \ln \left(\exp \left(-\frac{1}{2} \|x\|^2 \right) \right) \\ &= \ln \left((2\pi)^{-d/2} \right) - \frac{1}{2} \|x\|^2 \end{aligned}$$

By inspection the maximum of $L(x)$ is achieved at the mean, $x = 0$. Hence, the maximum density is:

$$(2\pi)^{-d/2}$$

$\therefore$ The maximum density for a spherical Gaussian $N(0, I_d)$ is $(2\pi)^{-d/2}$, and the point is achieved at $x = 0$

Solution 1

Solution 1 (b)

From *part (a)*, the maximum density is:

$$f_{\max} = (2\pi)^{-d/2}$$

Consider the limit of the density as d approaches infinity:

$$\lim_{d \rightarrow \infty} (2\pi)^{-d/2} = 0$$

$\therefore$ The maximum density approaches 0 as d grows.

Solution 2

Solution 2 (a)

Python Code: d=10

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # set dimension and number of samples
5 d = 10
6 num_samples = 10000
7 np.random.seed(42)
8
9 # get 10000 points from  $N(0, I_d)$ 
10 samples = np.random.normal(0, 1, size=(num_samples, d))
11
12 # calc the squared L2 norm for each point
13 squared_lengths_d10 = np.sum(samples**2, axis=1)
14
15 # calculate mean
16 sample_mean_d10 = np.mean(squared_lengths_d10)
17 print(f"Dimension (d): {d}")
18 print(f"Sample Mean of  $\|X\|^2$ : {sample_mean_d10:.4f}")
19
20 # plot the histogram
21 plt.figure(figsize=(10, 6))
22 plt.hist(squared_lengths_d10, bins=50, color='skyblue', edgecolor='black', alpha=0.7,
23         density=True, label=f'Histogram of {num_samples} samples')
24 plt.axvline(d, color='red', linestyle='--', linewidth=2,
25            label=f'Theoretical Mean (d = {d}):  $E[\|X\|^2]$ ')
26 plt.axvline(sample_mean_d10, color='blue', linestyle=':', linewidth=2,
27            label=f'Sample Mean = {sample_mean_d10:.4f}')
28
29 # set plot args and save
30 plt.title('Histogram of Squared Lengths for  $N(0, I_{10})$ ', fontsize=14)
31 plt.xlabel('Squared Length  $\|X\|^2$ ', fontsize=12)
32 plt.ylabel('Density', fontsize=12)
33 plt.legend()
34 plt.grid(True, linestyle='--', alpha=0.5)
35 plt.savefig('histogram_d10.png', dpi=150, bbox_inches='tight')
```

Solution 2

Solution 2 (a)

Plot

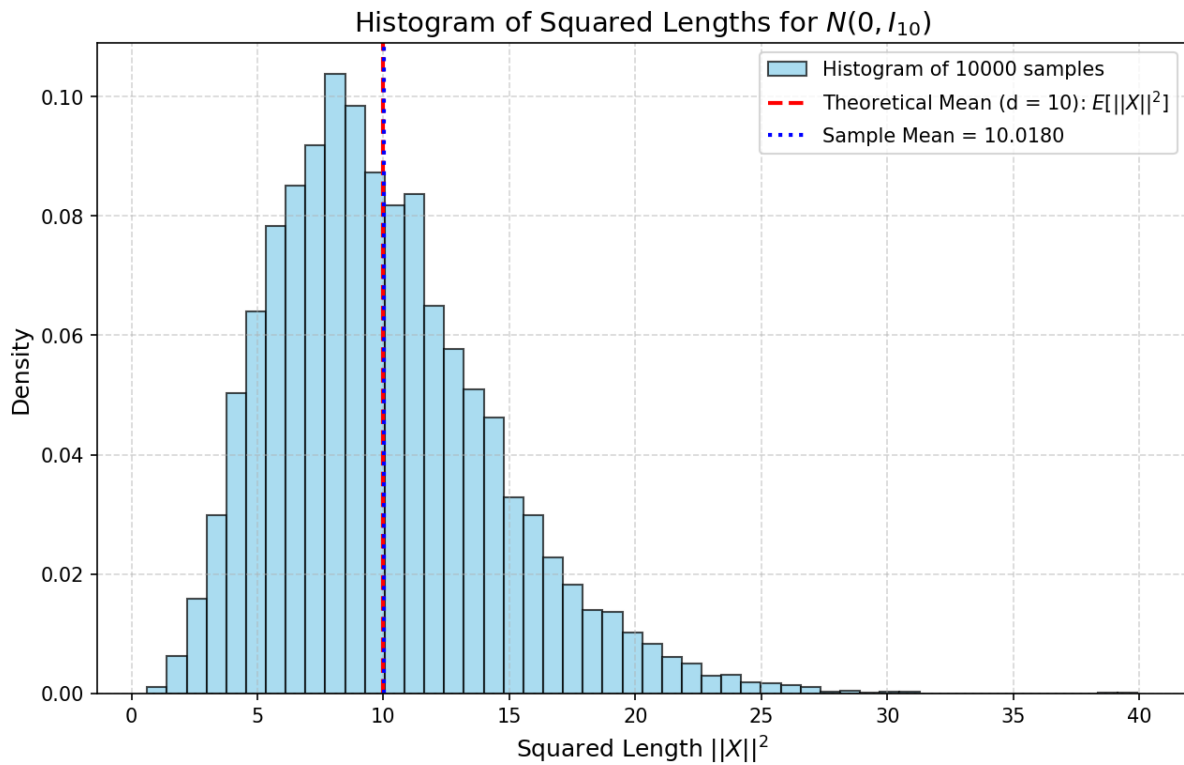


Figure 1: Histogram of squared lengths for $d=10$.

$\therefore$ The typical (squared) length is approximately 10

Solution 2

Solution 2 (b)

Python Code: d=100

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # set dimension and number of samples
5 d = 100
6 num_samples = 10000
7 np.random.seed(42)
8
9 # get 10000 points from  $N(0, I_d)$ 
10 samples = np.random.normal(0, 1, size=(num_samples, d))
11
12 # calc the squared L2 norm for each point
13 squared_lengths_d100 = np.sum(samples**2, axis=1)
14
15 # calculate mean
16 sample_mean_d100 = np.mean(squared_lengths_d100)
17 print(f"Dimension (d): {d}")
18 print(f"Sample Mean of  $\|X\|^2$ : {sample_mean_d100:.4f}")
19
20 # plot the histogram
21 plt.figure(figsize=(10, 6))
22 plt.hist(squared_lengths_d100, bins=50, color='skyblue', edgecolor='black', alpha=0.7,
23         density=True, label=f'Histogram of {num_samples} samples')
24 plt.axvline(d, color='red', linestyle='--', linewidth=2,
25            label=f'Theoretical Mean (d = {d}):  $E[\|X\|^2]$ ')
26 plt.axvline(sample_mean_d100, color='blue', linestyle=':', linewidth=2,
27            label=f'Sample Mean = {sample_mean_d100:.4f}')
28
29 # set plot args and save
30 plt.title('Histogram of Squared Lengths for  $N(0, I_{100})$ ', fontsize=14)
31 plt.xlabel('Squared Length  $\|X\|^2$ ', fontsize=12)
32 plt.ylabel('Density', fontsize=12)
33 plt.legend()
34 plt.grid(True, linestyle='--', alpha=0.5)
35 plt.savefig('histogram_d100.png', dpi=150, bbox_inches='tight')
```

Solution 2

Solution 2 (b)

Plot

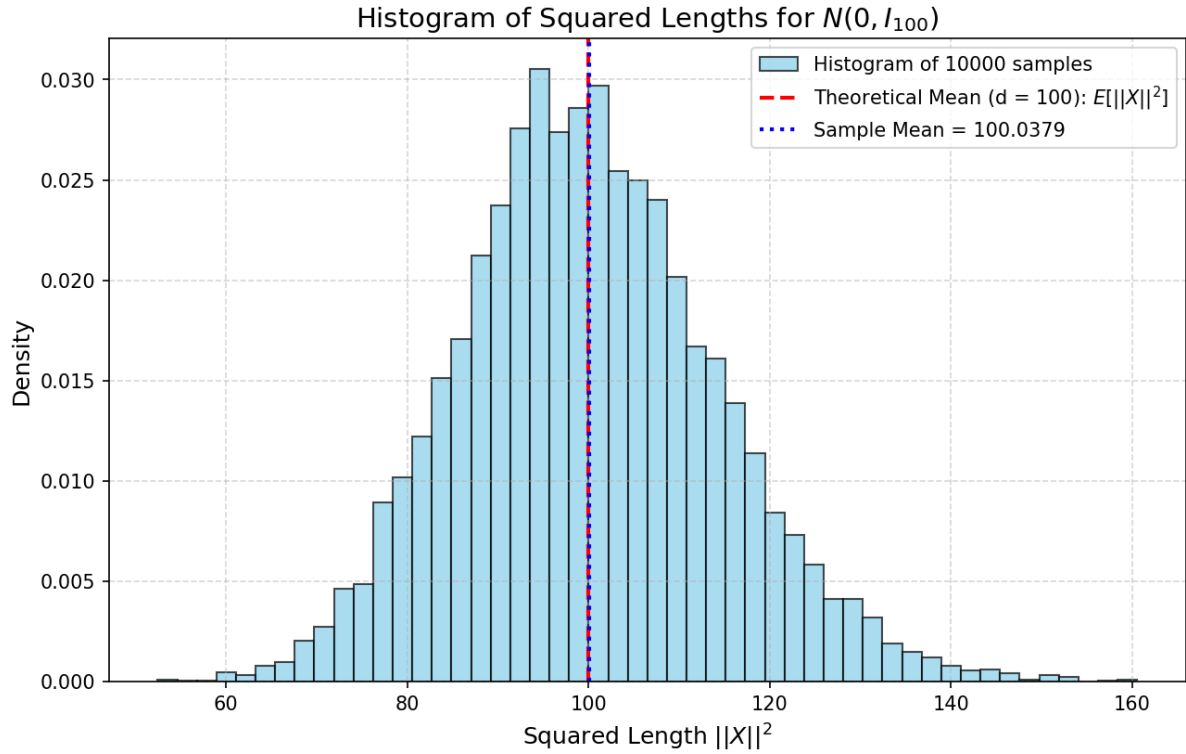


Figure 2: Histogram of squared lengths for $d=100$.

Calculate Concentration

To determine if the distribution is more concentrated, we must compare its relative spread. This is measured using the coefficient of variation or CV , which is defined as:

$$CV = \frac{\sigma}{\mu} \quad \text{where} \quad \mu = d, \sigma = \sqrt{2d}$$

For, $d = 10$ and $d = 100$, the coefficient of variation is:

$$CV_{10} = \frac{\sqrt{20}}{10} \approx 0.447 \quad \text{and} \quad CV_{100} = \frac{\sqrt{200}}{100} \approx 0.141$$

Hence, $CV_{100} < CV_{10}$

$\therefore$ the distribution for $d = 100$ is more concentrated than the distribution for $d = 10$

Solution 3

Let $X \sim N(\mu, \Sigma)$ where:

$$\mu = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \Sigma = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \quad v = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

1. Calculate the mean μ_Y :

$$\begin{aligned} \mu_Y &= v^\top \mu \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ &= \left(\frac{1}{\sqrt{2}} \cdot 1 \right) + \left(\frac{1}{\sqrt{2}} \cdot 2 \right) \\ &= \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{2}} \\ &= \frac{3}{\sqrt{2}} \end{aligned}$$

2. Calculate the variance σ_Y^2 :

$$\begin{aligned} \sigma_Y^2 &= v^\top \Sigma v \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2 \cdot (1/\sqrt{2}) + 0 \\ 0 + 4 \cdot (1/\sqrt{2}) \end{bmatrix} \\ &= [1/\sqrt{2} \quad 1/\sqrt{2}] \begin{bmatrix} 2/\sqrt{2} \\ 4/\sqrt{2} \end{bmatrix} \\ &= \left(\frac{1}{\sqrt{2}} \cdot \frac{2}{\sqrt{2}} \right) + \left(\frac{1}{\sqrt{2}} \cdot \frac{4}{\sqrt{2}} \right) \\ &= \frac{2}{2} + \frac{4}{2} \\ &= 1 + 2 = 3 \end{aligned}$$

$\therefore$ The distribution of the projection is $N\left(\frac{3}{\sqrt{2}}, 3\right)$

Solution 4

Solution 4 (a)

Let, $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N(\mu, \Sigma)$, where:

$$\mu = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Since the distribution for X_3 is a 1D Gaussian $N(\mu_3, \sigma_3^2)$, we can just lookup the mean and variance of X_3 from the given covariance matrix. Note, $\sigma_3^2 = \sigma_{(3,3)}^2$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \mu_3 = 0$$

$$\Sigma = \begin{bmatrix} \sigma_{(1,1)}^2 & \sigma_{(1,2)}^2 & \sigma_{(1,3)}^2 \\ \sigma_{(2,1)}^2 & \sigma_{(2,2)}^2 & \sigma_{(2,3)}^2 \\ \sigma_{(3,1)}^2 & \sigma_{(3,2)}^2 & \sigma_{(3,3)}^2 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix} \rightarrow \sigma_{(3,3)}^2 = 2$$

$\therefore$ The distribution of X_3 is:

$$N(0, 2)$$

Solution 4

Solution 4 (b)

Let, $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N(\mu, \Sigma)$, where:

$$\mu = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

Since we want the joint distribution for X_1 and X_2 , it follows that it is a 2D Gaussian $N(\mu_{1,2}, \Sigma_{1,2})$ or:

$$N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_{1,1}^2 & \sigma_{1,2}^2 \\ \sigma_{2,1}^2 & \sigma_{2,2}^2 \end{bmatrix}\right)$$

It follows that,

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \mu_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \mu_{1,2} = \mu_{1,2} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_{(1,1)}^2 & \sigma_{(1,2)}^2 & \sigma_{(1,3)}^2 \\ \sigma_{(2,1)}^2 & \sigma_{(2,2)}^2 & \sigma_{(2,3)}^2 \\ \sigma_{(3,1)}^2 & \sigma_{(3,2)}^2 & \sigma_{(3,3)}^2 \end{bmatrix} = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix} \rightarrow \Sigma_{1,2} = \begin{bmatrix} \sigma_{1,1}^2 & \sigma_{1,2}^2 \\ \sigma_{2,1}^2 & \sigma_{2,2}^2 \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}$$

$\therefore$ The joint distribution of X_1, X_2 is:

$$N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}\right)$$

Solution 4

Solution 4 (c)

Let, $X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N(\mu, \Sigma)$, where:

$$\mu = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

We need to find the conditional distribution of $X_1 \mid (X_2, X_3)$. Let $X_a = X_1$ and $X_b = \begin{bmatrix} X_2 \\ X_3 \end{bmatrix}$. We observe $x_b = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Now, partition the mean vector μ and covariance matrix Σ :

$$\mu = \begin{bmatrix} \mu_a \\ \mu_b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \text{and} \quad \Sigma = \begin{bmatrix} \Sigma_{aa} & \Sigma_{ab} \\ \Sigma_{ba} & \Sigma_{bb} \end{bmatrix} = \begin{bmatrix} 6 & 3 & 1 \\ 3 & 9 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

From this, we identify the blocks:

$$\mu_a = 0 \quad \mu_b = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \Sigma_{aa} = 6 \quad \Sigma_{ab} = \begin{bmatrix} 3 & 1 \end{bmatrix} \quad \Sigma_{ba} = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \quad \Sigma_{bb} = \begin{bmatrix} 9 & 0 \\ 0 & 2 \end{bmatrix}$$

The conditional distribution $X_a \mid X_b = x_b$ is also Gaussian, $N(\mu_{a|b}, \Sigma_{a|b})$, where:

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b) \quad \text{and} \quad \Sigma_{a|b} = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$$

1. Compute Σ_{bb}^{-1} : Since Σ_{bb} is diagonal, its inverse is the inverse of its diagonal elements:

$$\Sigma_{bb}^{-1} = \begin{bmatrix} 9 & 0 \\ 0 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix}$$

2. Compute the conditional mean $\mu_{a|b}$:

$$\mu_{a|b} = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(x_b - \mu_b) = 0 + \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix} \left(\begin{bmatrix} 2 \\ 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1/3 & 1/2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \frac{7}{6}$$

3. Compute the conditional variance $\Sigma_{a|b}$:

$$\Sigma_{a|b} = 6 - \begin{bmatrix} 3 & 1 \end{bmatrix} \begin{bmatrix} 1/9 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 6 - \begin{bmatrix} 1/3 & 1/2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 6 - \left(\left(\frac{1}{3} \cdot 3 \right) + \left(\frac{1}{2} \cdot 1 \right) \right) = \frac{9}{2}$$

$\therefore$ The resulting conditional distribution of X_1 is:

$$N\left(\frac{7}{6}, \frac{9}{2}\right)$$

Solution 5

Solution 5 (a)

A Gaussian Process is defined as a (potentially infinite) collection of random variables, any finite subset of which has a joint multivariate Gaussian distribution. The finite subset in this example is the 93 training and 11 test points, which we can express as:

$$\begin{bmatrix} y_{\text{tr}} \\ y_{\text{te}} \end{bmatrix}$$

$\therefore$ By the definition of a Gaussian Process, this finite collection of 104 random variables must have a joint multivariate Gaussian distribution

Solution 5

Solution 5 (b)

Python Code: Posterior Mean and MSE

```

1 import numpy as np
2 import pandas as pd
3 from scipy.spatial.distance import cdist
4
5 def load_data(filename):
6     data = pd.read_csv(filename, header=None)
7     data.columns = ['latitude', 'longitude', 'temperature']
8     X = data[['latitude', 'longitude']].values
9     y = data['temperature'].values
10    return X, y
11
12 def kernel(X1, X2, sigma=1.5):
13     # cdist computes the Euclidean distance between all pairs
14     dists = cdist(X1, X2, metric='euclidean')
15     return np.exp(-dists / sigma)
16
17 # load data
18 X_tr, y_tr = load_data('gptrain.csv')
19 X_te, y_te = load_data('gpctest.csv')
20 # set model parameters
21 SIGMA = 1.5
22 MEAN_VAL = 20.0
23 JITTER = 1e-6
24 # calc mean vectors
25 m_tr = np.full(y_tr.shape, MEAN_VAL)
26 m_te = np.full(y_te.shape, MEAN_VAL)
27 # calc kernel Matrices
28 K_tr = kernel(X_tr, X_tr, sigma=SIGMA)
29 K_tr_stable = K_tr + np.eye(K_tr.shape[0]) * JITTER
30 K_te_tr = kernel(X_te, X_tr, sigma=SIGMA)
31 # calc GP posterior mean
32 alpha = np.linalg.solve(K_tr_stable, y_tr - m_tr)
33 mu_post = m_te + K_te_tr @ alpha
34 # calc GP MSE
35 mse_gp = np.mean((mu_post - y_te)**2)
36 print(f"\nGP Posterior Mean: {mu_post}")
37 print(f"Actual Test Values: {y_te}")
38 print(f"GP Mean Squared Error (MSE): {mse_gp:.6f}")
39 # calc baseline MSE
40 baseline_pred = np.mean(y_tr)
41 mse_baseline = np.mean((baseline_pred - y_te)**2)
42 print(f"Baseline (Average Training) Prediction: {baseline_pred:.6f}")
43 print(f"Baseline Mean Squared Error (MSE): {mse_baseline:.6f}")

```

Code Output

```

GP Posterior Mean: [18.11251828 21.97092191 21.21172958 15.76871264 20.23993114 21.2800907
 16.96205493 21.84549615 20.6904639 20.2495184 18.08471762]
Actual Test Values: [16.3 22.1 20.4 14.8 19.7 20.5 18.4 19.1 22.6 22.9 17.4]
GP Mean Squared Error (MSE): 2.413174
Baseline (Average Training) Prediction: 18.970968
Baseline Mean Squared Error (MSE): 6.422837

```

∴ The Mean Squared Error of the Gaussian Process is 2.413174. The Mean Squared Error from predicting the average of the training data is 6.422837.

Solution 5

Solution 5 (c)

Python Code: Grid Predictions and Plots

```

1 import numpy as np
2 import pandas as pd
3 import matplotlib.pyplot as plt
4 from scipy.spatial.distance import cdist
5
6 def load_data(filename):
7     data = pd.read_csv(filename, header=None)
8     data.columns = ['latitude', 'longitude', 'temperature']
9     X = data[['latitude', 'longitude']].values
10    y = data['temperature'].values
11    return X, y
12
13 def kernel(X1, X2, sigma=1.5):
14     dists = cdist(X1, X2, metric='euclidean')
15     return np.exp(-dists / sigma)
16
17 # load data and set model parameters
18 X_tr, y_tr = load_data('gptrain.csv')
19 SIGMA = 1.5
20 MEAN_VAL = 20.0
21 JITTER = 1e-6
22
23 # pre-compute for posterior
24 m_tr = np.full(y_tr.shape, MEAN_VAL)
25 K_tr = kernel(X_tr, X_tr, sigma=SIGMA)
26 K_tr_stable = K_tr + np.eye(K_tr.shape[0]) * JITTER
27 alpha = np.linalg.solve(K_tr_stable, y_tr - m_tr)
28
29 # create grid (6400 points)
30 GRID_SIZE = 80
31 lat_min, lat_max = X_tr[:, 0].min(), X_tr[:, 0].max()
32 lon_min, lon_max = X_tr[:, 1].min(), X_tr[:, 1].max()
33 lat_grid = np.linspace(lat_min, lat_max, GRID_SIZE)
34 lon_grid = np.linspace(lon_min, lon_max, GRID_SIZE)
35 LON, LAT = np.meshgrid(lon_grid, lat_grid)
36
37 # stack data into (n_grid_points, 2) array
38 X_grid = np.column_stack([LAT.ravel(), LON.ravel()])
39 print(f"Created grid with {X_grid.shape[0]} points.")
40
41 # calc posterior mean on grid
42 m_grid = np.full(X_grid.shape[0], MEAN_VAL)
43 K_grid_tr = kernel(X_grid, X_tr, sigma=SIGMA)
44 mu_grid = m_grid + K_grid_tr @ alpha
45 mu_grid_reshaped = mu_grid.reshape(LAT.shape)
46
47 # calc posterior standard deviation on grid
48 K_diag = np.ones(X_grid.shape[0])
49
50 # use Cholesky decomposition and solve
51 L = np.linalg.cholesky(K_tr_stable)
52 V = np.linalg.solve(L, K_grid_tr.T)
53 diag_term = np.sum(V**2, axis=0)
54
55 # calc posterior variance
56 var_grid = K_diag - diag_term
57 # Clip at 0 for numerical stability before sqrt
58 std_grid = np.sqrt(np.maximum(var_grid, 0))
59 std_grid_reshaped = std_grid.reshape(LAT.shape)
60
61 # plot results
62 fig, ax = plt.subplots(1, 2, figsize=(16, 7))
63 plot_extent = [lon_min, lon_max, lat_min, lat_max]

```

```
64
65 # plot posterior mean
66 im1 = ax[0].imshow(mu_grid_reshaped, extent=plot_extent,
67                    origin='lower', aspect='auto', cmap='hot')
68 ax[0].scatter(X_tr[:, 1], X_tr[:, 0], c='blue', marker='x', label='Training Stations')
69 ax[0].set_xlabel('Longitude')
70 ax[0].set_ylabel('Latitude')
71 ax[0].set_title('GP Posterior Mean Temperature')
72 ax[0].legend()
73 fig.colorbar(im1, ax=ax[0], label='Temperature ($\deg$C)')
74
75 # plot posterior standard deviation
76 im2 = ax[1].imshow(std_grid_reshaped, extent=plot_extent,
77                    origin='lower', aspect='auto', cmap='viridis')
78 ax[1].scatter(X_tr[:, 1], X_tr[:, 0], c='red', marker='x', label='Training Stations')
79 ax[1].set_xlabel('Longitude')
80 ax[1].set_ylabel('Latitude')
81 ax[1].set_title('GP Posterior Standard Deviation')
82 ax[1].legend()
83 fig.colorbar(im2, ax=ax[1], label='Std. Deviation ($\deg$C)')
84
85 plt.tight_layout()
86 # Save the figure to be included in the LaTeX document
87 plt.savefig('gp_predictions.png', dpi=150, bbox_inches='tight')
```

Solution 5

Solution 5 (c)

Plots

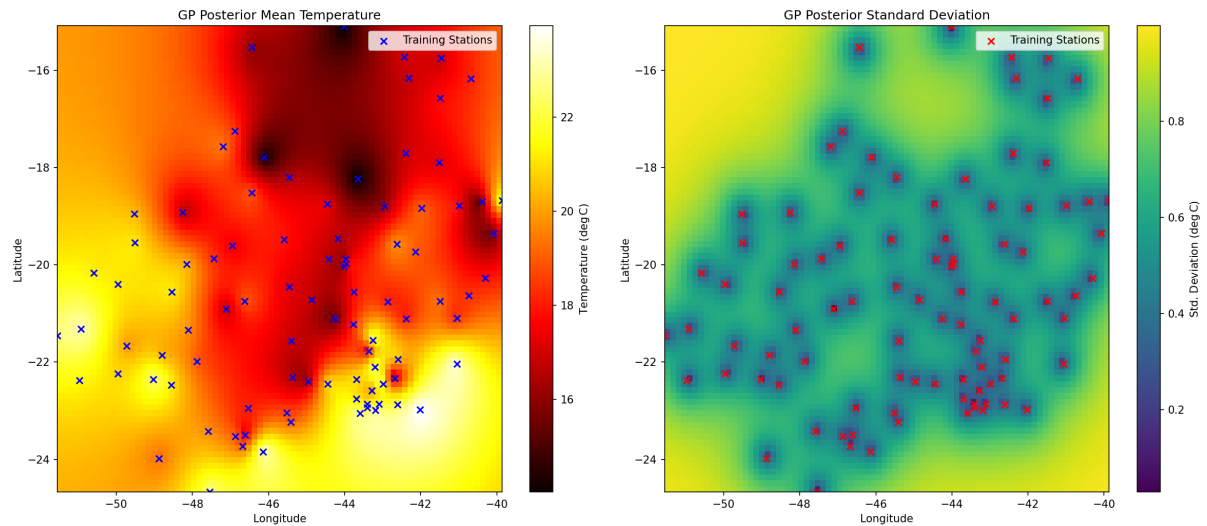


Figure 3: Posterior mean temperature (left) and posterior standard deviation (right) over the grid. Training station locations are marked with an 'x'.

Standard Deviation Relationship

The relationship is clear from the right-hand plot: The standard deviation of the prediction is lowest at or very close to the locations of the training stations. As the distance from the nearest training station increases, the standard deviation increases.

Solution 6

Solution 6 (a)

Construct Conditional Probability Tables (CPTs)

1. CPT for F_1 : $P(F_1)$

$P(F_1 = 1)$	$P(F_1 = 0)$
0.5	0.5

2. CPT for F_2 : $P(F_2)$

$P(F_2 = 1)$	$P(F_2 = 0)$
0.7	0.3

3. CPT for F_3 : $P(F_3 \mid F_1, F_2)$

F_1	F_2	$P(F_3 = 1)$	$P(F_3 = 0)$
0	0	0.3	0.7
0	1	0.8	0.2
1	0	0.8	0.2
1	1	0.8	0.2

4. CPT for F_4 : $P(F_4 \mid F_3)$

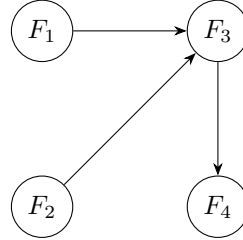
F_3	$P(F_4 = 1)$	$P(F_4 = 0)$
0	0.5	0.5
1	0.9	0.1

$\therefore$ The Conditional Probability Tables (CPTs) for the model can be seen above

Solution 6

Solution 6 (b)

Given:



The joint probability distribution specified by this Bayesian Network is:

$$P(F_1, F_2, F_3, F_4) = P(F_1) \cdot P(F_2) \cdot P(F_3 \mid F_1, F_2) \cdot P(F_4 \mid F_3)$$

From the tables in *part(a)*:

$$P(F_1 = 1) = 0.5 \quad P(F_2 = 1) = 0.7 \quad P(F_3 = 1 \mid F_1 = 1, F_2 = 1) = 0.8 \quad P(F_4 = 0 \mid F_3 = 1) = 0.1$$

Now, substitute the values into the equation above for $P(F_1, F_2, F_3, F_4)$:

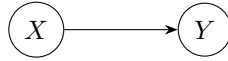
$$\begin{aligned}
 P(F_1 = 1, F_2 = 1, F_3 = 1, F_4 = 0) &= P(F_1 = 1) \times P(F_2 = 1) \times P(F_3 = 1 \mid F_1 = 1, F_2 = 1) \times P(F_4 = 0 \mid F_3 = 1) \\
 &= (0.5) \times (0.7) \times (0.8) \times (0.1) \\
 &= (0.35) \times (0.08) \\
 &= 0.028
 \end{aligned}$$

$\therefore$ The probability of the configuration $F_1 = F_2 = F_3 = 1, F_4 = 0$ is 0.028

Solution 7

Left DAG

Consider the DAG below:



The rules of Bayesian networks (that a variable is conditioned on its parents), the DAG above implies a factorization of the joint probability as:

$$P(X, Y) = P(X) \cdot P(Y | X)$$

From the chain rule of probability, any joint distribution $P(X, Y)$ can always be written as:

$$P(X)P(Y | X)$$

Hence, the factorization required by the Left DAG for any $P(X, Y)$

Right DAG

Consider the DAG below:



This DAG implies a factorization of the joint probability as:

$$P(X, Y) = P(Y) \cdot P(X | Y)$$

From the chain rule of probability, any joint distribution $P(X, Y)$ can always be written as:

$$P(Y)P(X | Y)$$

Hence, the factorization required by the DAG for any $P(X, Y)$

$\therefore$ Any joint distribution $P(X, Y)$ factors over both the DAGs

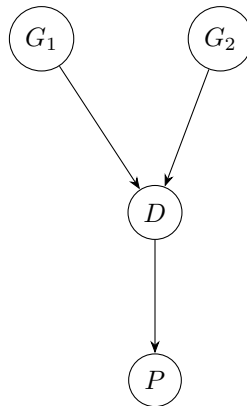
Solution 8

Solution 8 (a)

Bayes Net Diagram

Given:

- G_1 and G_2 are independent.
- D depends on G_1 and G_2 .
- P depends on D .



Construct Conditional Probability Tables (CPTs)

1. CPT for G_1 : $P(G_1)$

$P(G_1 = 1)$	$P(G_1 = 0)$
0.1	0.9

2. CPT for G_2 : $P(G_2)$

$P(G_2 = 1)$	$P(G_2 = 0)$
0.2	0.8

3. CPT for D : $P(D \mid G_1, G_2)$

G_1	G_2	$P(D = 1)$	$P(D = 0)$
0	0	0.0	1.0
0	1	0.1	0.9
1	0	0.2	0.8
1	1	0.5	0.5

4. CPT for P : $P(P \mid D)$

D	$P(P = 1)$	$P(P = 0)$
0	0.01	0.99
1	0.50	0.50

$\therefore$ The Bayes Net diagram and the conditional probability tables are seen above

Solution 8

Solution 8 (b)

From the conditional probability tables in *part (a)*:

$$\begin{aligned} P(G_1 = 0) &= 0.9 & P(D = 1 \mid G_1 = 0, G_2 = 0) &= 0.0 \\ P(G_2 = 0) &= 0.8 & P(D = 1 \mid G_1 = 0, G_2 = 1) &= 0.1 \\ P(G_1 = 1) &= 0.1 & P(D = 1 \mid G_1 = 1, G_2 = 0) &= 0.2 \\ P(G_2 = 1) &= 0.2 & P(D = 1 \mid G_1 = 1, G_2 = 1) &= 0.5 \end{aligned}$$

To find the marginal probability $P(D = 1)$, we have the equation below:

$$P(D = 1) = \sum_{g_1 \in \{0,1\}} \sum_{g_2 \in \{0,1\}} P(G_1 = g_1, G_2 = g_2, D = 1)$$

Expanding the summations the equation becomes:

$$\begin{aligned} P(D = 1) &= \sum_{g_1 \in \{0,1\}} \sum_{g_2 \in \{0,1\}} P(G_1 = g_1, G_2 = g_2, D = 1) \\ &= P(G_1 = 0, G_2 = 0, D = 1) \\ &\quad + P(G_1 = 0, G_2 = 1, D = 1) \\ &\quad + P(G_1 = 1, G_2 = 0, D = 1) \\ &\quad + P(G_1 = 1, G_2 = 1, D = 1) \end{aligned}$$

expand using: $P(G_1, G_2, D) = P(G_1)P(G_2)P(D \mid G_1, G_2)$

$$\begin{aligned} &= P(G_1 = 0)P(G_2 = 0)P(D = 1 \mid G_1 = 0, G_2 = 0) \\ &\quad + P(G_1 = 0)P(G_2 = 1)P(D = 1 \mid G_1 = 0, G_2 = 1) \\ &\quad + P(G_1 = 1)P(G_2 = 0)P(D = 1 \mid G_1 = 1, G_2 = 0) \\ &\quad + P(G_1 = 1)P(G_2 = 1)P(D = 1 \mid G_1 = 1, G_2 = 1) \end{aligned}$$

Now, substitute the values from the condition probability tables in *part (a)*:

$$\begin{aligned} P(D = 1) &= (0.9)(0.8)(0.0) \\ &\quad + (0.9)(0.2)(0.1) \\ &\quad + (0.1)(0.8)(0.2) \\ &\quad + (0.1)(0.2)(0.5) \\ &= 0.0 + 0.018 + 0.016 + 0.010 \\ &= 0.044 \end{aligned}$$

$\therefore$ The probability of disease ($D = 1$) is 0.044, or 4.4%

Solution 8

Solution 8 (c)

From *part (a)*:

$$P(P = 1 \mid D = 0) = 0.01 \quad \text{and} \quad P(P = 1 \mid D = 1) = 0.5$$

From *part (b)*:

$$P(D = 1) = 0.044 \rightarrow P(D = 0) = 1 - 0.044 = 0.956$$

To find the marginal probability $P(P = 1)$, we have the equation below:

$$P(P = 1) = \sum_{d \in \{0,1\}} P(D = d, P = 1)$$

We can rewrite this as:

$$P(P = 1) = P(P = 1 \mid D = 0)P(D = 0) + P(P = 1 \mid D = 1)P(D = 1)$$

Now, substitute the values from *part(a)* and *part(b)*:

$$\begin{aligned} P(P = 1) &= P(P = 1 \mid D = 0)P(D = 0) + P(P = 1 \mid D = 1)P(D = 1) \\ &= (0.01 \cdot 0.956) + (0.5 \cdot 0.044) \\ &= 0.03156 \end{aligned}$$

$\therefore$ The probability of paralysis ($P = 1$) is 0.03156, or approximately:

$$3.16\%$$

Solution 8

Solution 8 (d)

From *part (a)*:

$$P(P = 1 \mid D = 1) = 0.5$$

From *part (b)*:

$$P(D = 1) = 0.044$$

From *part (c)*:

$$P(P = 1) = 0.03156$$

Apply Bayes' Theorem to find the probability $P(D = 1 \mid P = 1)$:

$$P(D = 1 \mid P = 1) = \frac{P(P = 1 \mid D = 1)P(D = 1)}{P(P = 1)}$$

Now, substitute values from *part(a)*, *part(b)*, and *part(c)*

$$\begin{aligned} P(D = 1 \mid P = 1) &= \frac{P(D = 1 \mid P = 1, D = 1)P(D = 1)}{P(D = 1)} \\ &= \frac{0.5 \cdot 0.044}{0.03156} \\ &\approx 0.69708 \end{aligned}$$

$\therefore$ the approximate probability that a person with paralysis has the disease is:

$$P(D = 1 \mid P = 1) \approx 0.69708$$

Solution 8

Solution 8 (e)

From the conditional probability tables in *part (a)* and results from *part (b)*:

$$P(D = 1) = 0.044 \quad P(G_2 = 0) = 0.8$$

$$P(G_1 = 1) = 0.1 \quad P(D = 1 \mid G_1 = 1, G_2 = 0) = 0.2$$

$$P(G_2 = 1) = 0.2 \quad P(D = 1 \mid G_1 = 1, G_2 = 1) = 0.5$$

Apply Bayes' Theorem to find the probability to calculate $P(G_1 = 1 \mid D = 1)$

$$P(G_1 = 1 \mid D = 1) = \frac{P(D = 1, G_1 = 1)}{P(D = 1)}$$

We can rewrite the numerator as the following:

$$P(G_1 = 1 \mid D = 1) =$$

$$\frac{P(G_1 = 1)P(G_2 = 0)P(D = 1 \mid G_1 = 1, G_2 = 0) + P(G_1 = 1)P(G_2 = 1)P(D = 1 \mid G_1 = 1, G_2 = 1)}{P(D = 1)}$$

Now substitute the values from *part (a)* and *part (b)*:

$$\begin{aligned} P(G_1 = 1 \mid D = 1) &= \frac{(0.1)(0.8)(0.2) + (0.1)(0.2)(0.5)}{0.044} \\ &= \frac{0.016 + 0.010}{0.044} \\ &= \frac{0.026}{0.044} \\ &\approx 0.5909 \end{aligned}$$

$\therefore$ The probability that a person with the disease has the mutant G_1 gene is approximately:

$$P(G_1 = 1 \mid D = 1) \approx 0.591$$

Solution 9

Solution 9 (a)

Python Code

```
1 print(f"Total characters in vocabulary: {len(char_to_index)}")
2 print(f"Integer code for 'A': {char_to_index['A']}")
```

∴ There are 66 unique characters in the vocabulary. The integer code for 'A' is 12

Solution 9

Solution 9 (b)

Python Code

```
1 print(f"Number of characters in shakespeare_plays.txt: {len(data)}")
```

∴ There are 3801088 characters in `shakespeare_plays.txt`

Solution 9

Solution 9 (c)

The size 66 is used for both the input and output layers because it corresponds to the total number of unique characters in the vocabulary as found in *part (a)*.
